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Intelli Route: Smart Navigation System

Final Project Report

1. Introduction

Intelli Route is a smart navigation system designed to compute optimal routes between locations using advanced graph algorithms. The system leverages **Data Structures and Algorithms (DSA)** concepts to simulate real-world navigation, like GPS-based systems.

The project demonstrates how algorithms like **Dijkstra** and **A*** can efficiently solve shortest path problems while incorporating **traffic-aware routing** to improve decision-making.

2. Problem Statement

In real-world navigation systems:

- Multiple possible routes exist between locations, making it difficult to identify the most efficient path
- Traffic conditions dynamically change route efficiency and travel time
- Traditional static routing methods fail to adapt to real-time conditions
- There is a need for **fast, accurate, and optimal pathfinding solutions**

3. System Design & Architecture

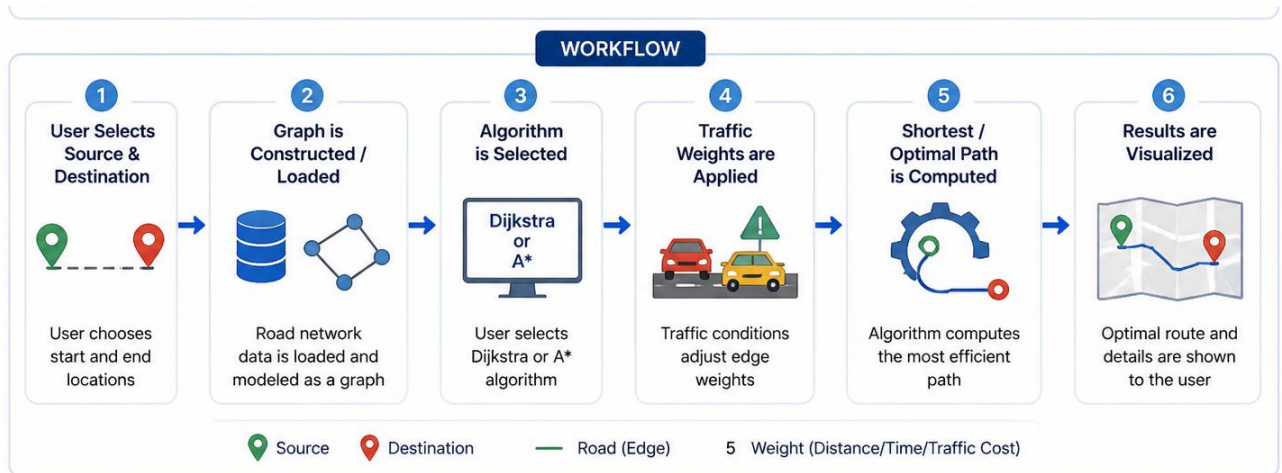
The system models a road network as a **graph**, where:

- Nodes represent locations (cities/intersections)

- Edges represent roads
- Edge weights represent distance, time, or traffic cost

Workflow:

1. User selects source and destination
2. Graph is constructed/loaded
3. Algorithm (Dijkstra or A*) is selected
4. Traffic weights are applied
5. Shortest/optimal path is computed
6. Results are visualized



4. Algorithms Used

4.1 Dijkstra's Algorithm

- Finds the shortest path in weighted graphs
- Guarantees optimal solution
- Explores all possible paths systematically

Time Complexity:

$$O((V + E) \log V)$$

4.2 A* Algorithm

- Uses heuristic (Euclidean distance) to guide search

- Faster than Dijkstra in most cases
- Reduces unnecessary node exploration

Time Complexity:

$O(E \log V)$

5. Data Structures Used

Data Structure	Purpose
Graph (Adjacency List)	Represents the road network
Priority Queue (Min-Heap)	Selects node with minimum cost
Hash Map (Dictionary)	Stores distances and visited nodes
Set	Tracks explored nodes
List / Array	Stores final path

6. Traffic Simulation

The system incorporates a basic traffic model by:

- Adjusting edge weights dynamically
- Simulating congestion levels
- Modifying route cost based on traffic intensity

This allows the system to mimic **real-world traffic-aware navigation**.

7. Implementation

The project includes:

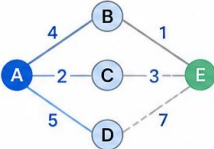
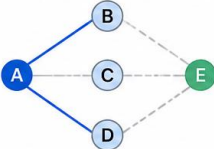
- **Web App (Intelliroute.html)** for visualization
- Live server for algorithm testing

- Interactive UI with:
 - Route visualization
 - Algorithm comparison
 - Traffic simulation

8. Results & Analysis

- A* algorithm showed **faster execution time** compared to Dijkstra
- A* explored fewer nodes due to heuristic guidance
- Both algorithms produced **optimal shortest paths**
- Traffic simulation improved route decision realism

9. Complexity Analysis

Algorithm	Time Complexity	Space Complexity
 Dijkstra's Algorithm Finds shortest path using min-heap	 $O((V + E) \log V)$ Uses priority queue (min-heap) to select the minimum distance node	 $O(V)$ Stores distances, visited nodes and other auxiliary data
 A* Algorithm Uses heuristic to find optimal path faster	 $O(E \log V)$ Heuristic reduces the number of explored nodes, making it faster in practice	 $O(V)$ Stores open set, closed set, heuristic values and other data
V V = Number of vertices (nodes) E E = Number of edges	 Time Complexity shows how running time increases with larger inputs  Space Complexity shows how memory usage increases with larger inputs	

10. Conclusion

This project demonstrates a real-world application of **data structures and algorithms** in building an intelligent navigation system.

Key takeaways:

- Efficient pathfinding using graph algorithms
- Improved performance using heuristics (A*)
- Scalable system design using appropriate data structures
- Real-world relevance through traffic-aware routing

11. Future Scope

- Integration with real-time APIs (Google Maps, traffic data)
- Machine learning for predictive routing
- Multi-modal transportation (bus, train, EV routing)
- Live GPS tracking

12. How to Run the Project- IntelliRoute System

Step 1: Extract the Project Files

Unzip the project folder named **smart_route_planner** to your local system.

Step 2: Open the Application

Inside the extracted folder, locate the file **IntelliRoute.html**.

Right-click on the file and select:

Open with → **Live Server** (recommended for proper rendering).

Step 3: Launch the Interface

Once opened, the IntelliRoute web interface will load in your browser.

Step 4: Use the Control Panel

On the left-hand side, you will find the **Control Panel**, where you can:

- Enter the **Source location**
- Select the **Destination**
- Choose the **Optimization mode** (e.g., shortest path)
- Select the **Algorithm** (A*, Dijkstra, or both)

Step 5: Explore Features

Use the navigation tabs at the top of the interface to explore:

- **Planner** – Route computation and visualization
- **Algorithms** – Step-by-step execution
- **Traffic** – Traffic simulation insights
- **Analysis** – Performance and route comparison

Step 6: View Results

Click on “**Find Best Route**” to compute and visualize the optimal path along with distance, time, and other metrics.

13. References

- Dijkstra, E. W. (1959). A note on two problems in connection with graphs
- A* Algorithm (Hart, Nilsson, Raphael, 1968)
- Standard Data Structures textbooks